GLA UNIVERSITY, MATHURA

End Term Examination, Even Semester 2025–26

B.Tech. CSE(DA), III Year (VI Semester)

BCSE0556 – Hadoop and Big Data Analytics

Course Outcome

CO1: Understand the concepts and challenges of big data.

CO2: Apply existing technology to collect, manage, store, query, and analyze the big data.

CO3: Apply job scheduling and resource management using Hadoop and Yarn.

CO4: Apply data summarization, query, and analysis of big data using Pig and Hive.

CO5: Design the regression model, cluster and decision tree of big data.

CO6: Experiment with hands-on experience in large-scale analytics tools to solve big data problems.

Printed Pages: 04 **SET B** University Roll No. ....

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. All questions should be answered into given sequenced order for all sections.
2. Answer should be brief and to-the-point with neat sketch/diagram.
3. Any missing or wrong data may be assumed to be suitable with giving proper justification.
4. Mentioned on the right-hand side margin indicates the full marks for respective questions.

Section – A

Attempt All Questions **4 X 5 = 20 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss various types of digital data. Also describe the characteristics of data (5 V's: Volume, Velocity, Variety, Veracity, Value) with one real-world example for each.	4	1	R	C
2	What do you understand by Hadoop Distributed File System (HDFS)? Analyze its importance in Hadoop and explain the roles of NameNode, DataNode, and Secondary NameNode.	4	3	U	M
3	How is Apache Spark different from MapReduce? Also list the key features of Spark and explain RDD (Resilient Distributed Dataset) and DAG execution model.	4	3	An	C
4	State the usage of *'filter'*, *'group'*, *'orderBy'*, *'distinct'* keywords in pig scripts. Give one example Pig Latin statement for each keyword.	4	2	A	F
5	What kind of applications are supported by Apache Hive? Is Hive designed for OLAP? Justify your answer. **Or** Write a command for the following in Hive Query Language: i) Changing directory to HIVE_HOME ii) Creating a database named university_2026 iii) Creating a table student with fields: rollno, name, branch, cgpa iv) Loading data from file at /root/data/students.csv v) Counting number of rows in the table vi) Exiting the Hive CLI	4	4	E	P

Section – B

Attempt All Questions **3 X 5 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
6	Define: a) Atom b) Tuple c) Field d) Bag e) Relation f) Show in diagrammatic representation for above.	3	6	R	F
7	What is HBase? Illustrate the key components of HBase and explain Column Families, Rows, Cells, and Timestamps in HBase data model.	3	1	R	F
8	Write the complete steps to import a CSV file into Apache Cassandra. Also write CQL to export data from a Cassandra table back to a CSV file.	3	2	U	P
9	Why partitioning is used in Hive? Define and differentiate static and dynamic partitioning in Hive. Write HiveQL to create a partitioned table by department and load data into it.	3	1	R	F
10	Does Pig differ from MapReduce? If yes, how? Explain with reference to abstraction level, programming model, and use-cases.	3	1	An	C

Section – C

Attempt All Questions **5 X 3 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
11	Explain CRUD operations in Cassandra with suitable CQL examples. Also explain how Cassandra handles distributed writes using Partitioner, Replication Factor, and Consistency Level. Draw the ring architecture.	5	2	E	PC
12	Explain the various components of Hadoop Ecosystem. Discuss HDFS, MapReduce, YARN, Pig, Hive, HBase, Sqoop, Zookeeper, and Kafka with a complete ecosystem diagram.	5	5	U	C
13	Explain the execution modes of Apache Pig in detail: a) Local Mode vs MapReduce Mode vs Tez Mode — differences with use-cases b) Interactive Mode (Grunt Shell) — how to run Pig Latin interactively c) Batch Mode — how to run a .pig script file d) Write a Pig Latin script to load sales.csv, filter records where quantity > 100, group by product_category, compute total revenue per category, and store output to HDFS.	5	6	R	P